

Practical 1

A. Implement FCFS scheduling algorithm in Python.

Code :

```
def get_positive(msg):
    while True:
        try:
            x = int(input(msg))
            if x > 0:
                return x
            print("Invalid input! Enter a positive number.")
        except ValueError:
            print("Invalid input! Enter numbers only.")
n = get_positive("Enter number of processes: ")
bt = []
for i in range(n):
    bt.append(get_positive("Enter Burst Time for P" + str(i + 1) + ": "))
wt = [0]
tat = []
for i in range(1, n):
    wt.append(wt[i-1] + bt[i-1])
for i in range(n):
    tat.append(wt[i] + bt[i])
total_wt = 0
total_tat = 0
print("\nProcess\tBT\tWT\tTAT")
for i in range(n):
    print("P" + str(i+1), "\t", bt[i], "\t", wt[i], "\t", tat[i])
    total_wt += wt[i]
    total_tat += tat[i]
print("\nAverage Waiting Time =", total_wt/n)
print("Average Turnaround Time =", total_tat/n)
```

FCFS Output :

```
Enter number of processes: 2
Enter Burst Time for P1: 3
Enter Burst Time for P2: 4

Process BT      WT      TAT
P1      3      0      3
P2      4      3      7

Average Waiting Time = 1.5
Average Turnaround Time = 5.0
```

B. Implement SJF (with no preemption) scheduling algorithm in Python

Code :

```

def get_positive(msg):
    while True:
        try:
            x = int(input(msg))
            if x > 0:
                return x
            print("Invalid input! Enter a positive number.")
        except ValueError:
            print("Invalid input! Enter numbers only.")
n = get_positive("Enter number of processes: ")
bt = []
process = []
for i in range(n):
    process.append(i+1)
    bt.append(get_positive("Enter Burst Time for P" + str(i+1) + ": "))
for i in range(n):
    for j in range(i+1, n):
        if bt[i] > bt[j]:
            bt[i], bt[j] = bt[j], bt[i]
            process[i], process[j] = process[j], process[i]
wt = [0]
tat = []
for i in range(1, n):
    wt.append(wt[i-1] + bt[i-1])
total_wt = 0
total_tat = 0
print("\nProcess\tBT\tWT\tTAT")
for i in range(n):
    tat.append(wt[i] + bt[i])
    print("P" + str(process[i]), "\t", bt[i], "\t", wt[i], "\t", tat[i])
    total_wt += wt[i]
    total_tat += tat[i]
print("\nAverage Waiting Time =", total_wt/n)
print("Average Turnaround Time =", total_tat/n)

```

Output :

```

Enter number of processes: 2
Enter Burst Time for P1: 3
Enter Burst Time for P2: 4

```

Process	BT	WT	TAT
P1	3	0	3
P2	4	3	7

```

Average Waiting Time = 1.5
Average Turnaround Time = 5.0

```